Ayrık Zamanlı Fourier Transformu (Dönüşümü)

(Discrete-time Fourier Transform)

FT, periyodik olmayan işaretler için tanımlıdır.

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]\, e^{-j\omega n}$$

$$x[n] = \frac{1}{2\pi} \int_{2\pi} X(e^{j\omega}) e^{j\omega n}\, d\omega$$

$$x[n] \xrightarrow{FF} X(e^{j\omega}), \qquad FF^{-1}$$

⇒ Ayrık zamanlı FT, $X(e^{j\omega})$ 2π ile periyodiktir.

$$X(e^{j\omega}) = X\left(e^{j(\omega+2\pi)}\right)$$

Ex: $x[n] = a^n u[n]$, $|a| < 1$ ise $X(e^{j\omega}) = ?$

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n} = \sum_{n=0}^{\infty} a^n e^{-j\omega n} = \sum_{n=0} (ae^{-j\omega})^n = \frac{1}{1-ae^{-j\omega}}$$

($u[n] = 1,\ n \geq 0$)

$$\sum_{k=0}^{\infty} \alpha^k = \frac{1}{1-\alpha}, \quad |\alpha| < 1$$

$$a^n u[n] \xleftrightarrow{FF} \frac{1}{1-ae^{-j\omega}}$$

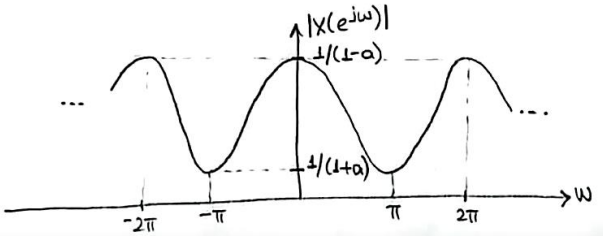
$$X(e^{j\omega}) = |X(e^{j\omega})|\, e^{j \measuredangle X(e^{j\omega})}$$

$$X(e^{j\omega}) = \frac{1}{1-ae^{-j\omega}} = \frac{1}{1-a\cos\omega + ja\sin\omega}$$

($e^{-j\omega} = \cos\omega - j\sin\omega$)

$$|X(e^{j\omega})| = \frac{1}{\sqrt{(1-a\cos\omega)^2 + a^2\sin^2\omega}} = \frac{1}{\sqrt{1+a^2-2a\cos\omega}}$$

$$\measuredangle X(e^{j\omega}) = -a\tan\left(\frac{a\sin\omega}{1-a\cos\omega}\right)$$



Ex: $x[n] = a^{|n|}$, $|a|<1$ ise $X(e^{j\omega}) = ?$

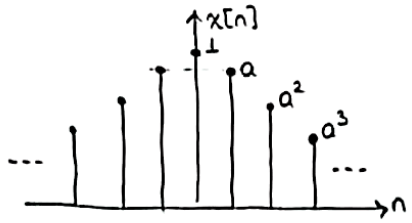
$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n} = \sum_{n=-\infty}^{\infty} a^{|n|}e^{-j\omega n} = \sum_{n=-\infty}^{-1} a^{n}e^{-j\omega n} + \sum_{n=0}^{\infty} a^{n}e^{-j\omega n}$$

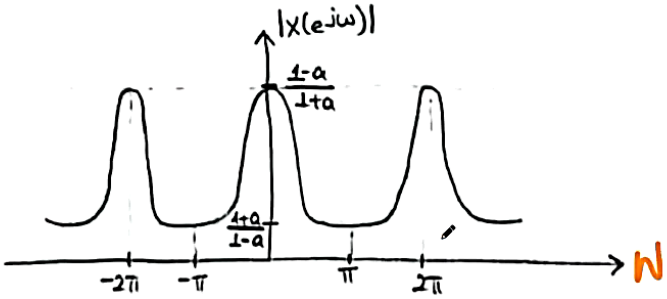
$$\sum_{n=-\infty}^{-1} (ae^{j\omega})^{-n} = \sum_{k=1}^{\infty} (ae^{j\omega})^{k} = \sum_{k=0}^{\infty} (ae^{j\omega})^{k} - 1 = \frac{1}{1-ae^{+j\omega}} - 1 = \frac{ae^{j\omega}}{1-ae^{j\omega}}$$

($k=-n$; $(ae^{j\omega})^0 = 1$)

$$X(e^{j\omega}) = \frac{ae^{j\omega}}{1-ae^{j\omega}} + \frac{1}{1-ae^{-j\omega}} = \frac{1-a^2}{1+a^2-2a\cos\omega} \quad : \text{Reel} \qquad X(e^{j\omega}) = |X(e^{j\omega})|$$

$$\measuredangle X(e^{j\omega}) = 0$$





$$a^{|n|} \xleftrightarrow{FT} \frac{1-a^2}{1+a^2-2a\cos\omega}$$

Ex: $x[n] = \begin{cases} 1, & |n| \le N_1 \\ 0, & |n| > N_1 \end{cases}$ şeklinde verilen ayrık zamanlı dikdörtgen darbe işaretinin FT bulunuz.

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n} = \sum_{n=-N_1}^{N_1} e^{-j\omega n} = \sum_{n=0}^{2N_1} e^{-j\omega(n-N_1)} = e^{j\omega N_1}\sum_{n=0}^{2N_1} e^{-j\omega n}$$

$$= e^{j\omega N_1}\left[\frac{1-e^{-j\omega(2N_1+1)}}{1-e^{-j\omega}}\right]$$

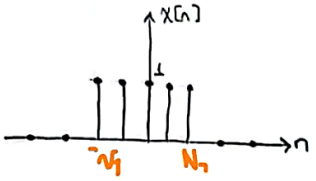
Reminder: $\sum_{k=0}^{M} \alpha^k = \frac{1-\alpha^{M+1}}{1-\alpha}$

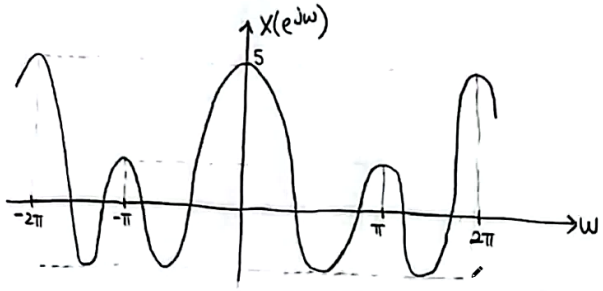
$$1-e^{j\theta} = e^{-j\theta/2}\left[e^{j\theta/2} - e^{-j\theta/2}\right]$$

$$= e^{j\omega N_1}\,\frac{e^{-j\omega(N_1+1/2)}\left[e^{j\omega(N_1+1/2)} - e^{-j\omega(N_1+1/2)}\right]}{e^{-j\omega/2}\left[e^{j\omega/2} - e^{-j\omega/2}\right]}$$

$$= \frac{\sin(\omega(N_1+1/2))}{\sin(\omega/2)}$$

$N_1 = 2$ için





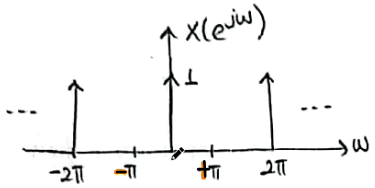
Ex: $x[n] = \delta[n]$

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n} = \sum_{n=-\infty}^{\infty} \delta[n] e^{-j\omega n} = e^{-j\omega 0} = 1$$

$\delta[n] = 1,\ n = 0$

Ex: $X(e^{j\omega}) = \delta(e^{j\omega})$ ise $x[n] = ?$

$$x[n] = \frac{1}{2\pi} \int_{2\pi} X(e^{j\omega}) e^{j\omega n} d\omega = \frac{1}{2\pi} \int_{-\pi}^{\pi} \delta(e^{j\omega}) e^{j\omega n} d\omega = \frac{1}{2\pi} \int_{-\pi}^{\pi} \delta(e^{j\omega}) e^{j\omega 0} d\omega = \frac{1}{2\pi}$$



$$\delta[n] \xleftrightarrow{FF} 1$$

$$1 \longleftrightarrow 2\pi\, \delta(e^{j\omega})$$

Ex: $x[n] = \delta[n - n_0]$

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n} = \sum_{n=-\infty}^{\infty} \delta[n-n_0] e^{-j\omega n} = e^{-j\omega n_0}$$

$\delta[n-n_0] = 1,\ n = n_0$

$$\delta[n-n_0] \longleftrightarrow e^{-j\omega n_0}$$

Ex: $X(e^{j\omega}) = \delta(e^{j(\omega-\omega_0)})$ ise $x[n] = ?$

$$x[n] = \frac{1}{2\pi} \int_{2\pi} X(e^{j\omega}) e^{j\omega n} d\omega = \frac{1}{2\pi} \int_{-\pi}^{\pi} \delta(e^{j(\omega-\omega_0)}) e^{j\omega n} d\omega$$

$$= \frac{1}{2\pi} \int_{2\pi} \delta(e^{j(\omega-\omega_0)}) e^{j\omega n} d\omega$$

$$= \frac{1}{2\pi} e^{j\omega_0 n} \underbrace{\int_{-\pi}^{\pi} \delta(e^{j(\omega-\omega_0)}) d\omega}_{=1} = \frac{1}{2\pi} e^{j\omega_0 n}$$

$$e^{j\omega_0 n} \xleftrightarrow{FF} 2\pi\, \delta(e^{j(\omega-\omega_0)})$$

$$e^{-j\omega_0 n} \xleftrightarrow{FF} 2\pi\, \delta(e^{j(\omega+\omega_0)})$$

Periyodik işaretlerin Fourier Transformu

$x[n] = x[n+N]$, $\forall n$, $N_0 = \frac{2\pi}{\omega_0}$, $\omega_0 = \frac{2\pi}{N}$

$$x[n] = \sum_{k=<N>} x_k e^{jk\omega_0 n} = x_0 + x_1 e^{j\omega_0 n} + x_2 e^{j2\omega_0 n} + \cdots + x_{N-1} e^{j(N-1)\omega_0 n}$$

$$x_k = \frac{1}{N} \sum_{n=<N>} x[n] e^{-jk\omega_0 n} \quad , \quad x_k = x_{k+N} \longleftarrow \begin{array}{l} x_0 = x_N \\ x_1 = x_{N+1} \\ \vdots \end{array}$$

$$x[n] = \sum_{k=<N>} x_k e^{jk\omega_0 n} \xleftrightarrow{FF} X(e^{j\omega}) = \sum_{k=-\infty}^{\infty} 2\pi x_k \delta(\omega - k\omega_0)$$

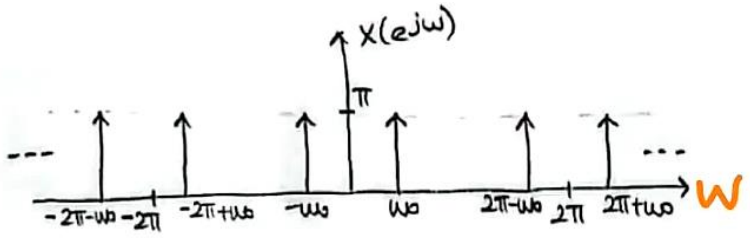
Ex: $x[n] = \cos\left(\frac{2\pi n}{5}\right)$ ise $X(e^{j\omega}) = ?$

$$x[n] = \frac{1}{2} e^{j(2\pi n/5)} + \frac{1}{2} e^{-j(2\pi n/5)} \quad , \quad \omega_0 = 2\pi/5$$

$$x_1 = x_{-1} = 1/2$$

$\updownarrow$ FF

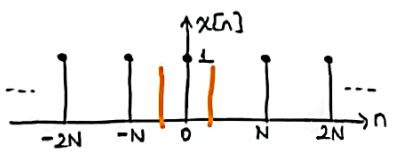
$$X(e^{j\omega}) = \pi\delta(\omega - 2\pi/5) + \pi\delta(\omega + 2\pi/5) \quad , \quad -\pi \le \omega < \pi$$

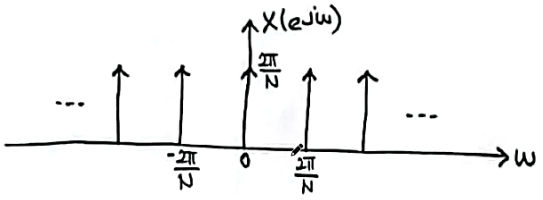


Ex: $x[n] = \sum_{k=-\infty}^{\infty} \delta[n - kN]$ periyodik dürtü treninin FT bulunuz.

Reminder: $x_k = \frac{1}{N} \sum_{n=<N>} x[n] e^{-jk\omega_0 n} = \frac{1}{N}$

$$X(e^{j\omega}) = \frac{2\pi}{N} \sum_{k=-\infty}^{\infty} \delta\left(\omega - \frac{2\pi k}{N}\right)$$





Ayrık Zamanlı Fourier Transformunun Özellikleri

$x[n] \xleftrightarrow{\mathcal{F}} X(e^{j\omega})$

$X(e^{j\omega}) = \mathcal{F}\{x[n]\}$

$x[n] = \mathcal{F}^{-1}\{X(e^{j\omega})\}$

1) Periyodiklik: $X(e^{j\omega}) = X(e^{j(\omega+2\pi)})$, 2π ile periyodiktir.

2) Lineerlik: $\alpha x_1[n] + \beta x_2[n] \xleftrightarrow{\mathcal{F}} \alpha X_1(e^{j\omega}) + \beta X_2(e^{j\omega})$

3) Zamanda öteleme: $x[n-n_0] \xleftrightarrow{\mathcal{F}} e^{-j\omega n_0} X(e^{j\omega})$

Genlik spektrumu değişmez, faz spektrumu $-\omega n_0$ kadar ötelenir.

4) Frekansta öteleme: $e^{j\omega_0 n} x[n] \xleftrightarrow{\mathcal{F}} X(e^{j(\omega-\omega_0)})$

5) Zamanda katlama: $x[-n] \xleftrightarrow{\mathcal{F}} X(e^{-j\omega})$

6) Eşlenik alma: $x^*[n] \xleftrightarrow{\mathcal{F}} X^*(e^{-j\omega})$

7) Fark alma (differencing): $x[n] - x[n-1] \xleftrightarrow{\mathcal{F}} (1 - e^{-j\omega}) X(e^{j\omega})$

8) Birikim (Accumulation): $\sum_{m=-\infty}^{n} x[m] \xleftrightarrow{\mathcal{F}} \frac{1}{1-e^{-j\omega}} X(e^{j\omega}) + \pi X(e^{j0}) \sum_{k=-\infty}^{\infty} \delta(\omega - 2\pi k)$

9) Frekansta türev: $n\,x[n] \xleftrightarrow{\mathcal{F}} j\frac{dX(e^{j\omega})}{d\omega}$

İspat: $X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$

$$\frac{d}{d\omega} X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] \frac{d}{d\omega}(e^{-j\omega n}) = -jn \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$$

10) Parseval teoremi: $E_x = \sum_{n=-\infty}^{\infty} |x[n]|^2 = \frac{1}{2\pi} \int_{2\pi} |X(e^{j\omega})|^2 d\omega$

11) Çarpma: $x_1[n] \cdot x_2[n] \xleftrightarrow{\mathcal{F}} \frac{1}{2\pi} \int_{2\pi} X_1(e^{j\alpha}) X_2(e^{j(\omega-\alpha)})\, d\alpha$

12) Konvolüsyon: $x_1[n] * x_2[n] \xleftrightarrow{\mathcal{F}} X_1(e^{j\omega}) X_2(e^{j\omega})$

$x[n] \longrightarrow \boxed{h[n]} \longrightarrow y[n]$

$y[n] = x[n] * h[n] \xleftrightarrow{\mathcal{F}} Y(e^{j\omega}) = X(e^{j\omega}) H(e^{j\omega})$

$h[n]$: Dürtü cevabı — $H(e^{j\omega})$: Frekans cevabı

$H(e^{j\omega}) = \frac{Y(e^{j\omega})}{X(e^{j\omega})} \xleftrightarrow{\mathcal{F}^{-1}} h[n]$

Ex: $x[n]=u[n]$ ise $X(e^{j\omega})=?$

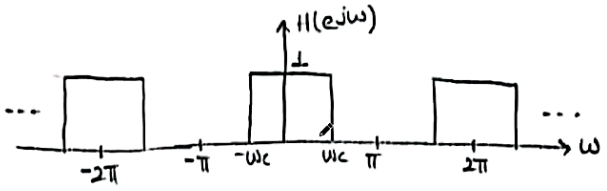
Reminder: $u[n] = \sum_{k=-\infty}^{n} \delta[k] = \sum_{k=0}^{\infty} \delta[n-k]$

$$\sum_{m=-\infty}^{n} x[m] \xleftrightarrow{\mathcal{F}} \frac{1}{1-e^{-j\omega}} X(e^{j\omega}) + \pi X(e^{j0}) \sum_{k=-\infty}^{\infty} \delta(\omega - 2\pi k)$$

$$\delta[n] \xleftrightarrow{\mathcal{F}} 1$$

$$u[n] \xleftrightarrow{\mathcal{F}} \frac{1}{1-e^{-j\omega}} + \pi \sum_{k=-\infty}^{\infty} \delta(\omega - 2\pi k)$$

Ex: $H(e^{j\omega}) = \begin{cases} 1, & 0 \le |\omega| \le \omega_c \\ 0, & \omega_c < |\omega| < \pi \end{cases}$ ayrık zamanlı alçak geçiren filtrenin dürtü cevabını bulunuz.



$$h[n] = \frac{1}{2\pi} \int_{2\pi} H(e^{j\omega}) e^{+j\omega n} d\omega = \frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} e^{j\omega n} d\omega = \frac{1}{2\pi} \frac{e^{j\omega n}}{jn} \Big|_{-\omega_c}^{\omega_c} = \frac{1}{2j\pi n} \left[e^{j\omega_c n} - e^{-j\omega_c n} \right]$$

$$h[n] = \frac{1}{\pi n} \sin(\omega_c n) = \frac{\sin\left(\pi \frac{\omega_c}{\pi} n\right)}{\pi \frac{\omega_c}{\pi} n} \cdot \frac{\omega_c}{\pi} = \frac{\omega_c}{\pi} \operatorname{sinc}\left(\frac{\omega_c}{\pi} n\right)$$

